Perpendicular edges and facet capacity in compact right-angled hyperbolic four-polytopes

Coxeter Polytopes Frontier Project

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Abstract

If a facet of a compact right-angled hyperbolic four-polytope has V vertices, the polytope has at least $2V + 5$ facets. The proof counts the opposite endpoints of edges perpendicular to a chosen facet, then uses an ear argument to control the remaining facets. Equality in the three-dimensional local count has the combinatorics of a Löbell polyhedron; the disjointness of its opposite face supplies the needed four-dimensional capacity estimate. Together with the exact pentagon excess identity, the result shows that a polytope of volume strictly below that of the right-angled 120-cell must have at most 48 vertices in every facet. It does not establish the 120-cell minimum or a second volume gap.

1 Statement and scope

Throughout, a polytope is convex and full-dimensional. A right-angled hyperbolic polytope has dihedral angle $\pi/2$ at every ridge. All polytopes in this paper are compact. In particular every edge has two finite endpoints, and every two-face is a compact right-angled polygon. If $P \subset \mathbb{H}^4$, write (v, e, r, f) for its numbers of vertices, edges, ridges and facets.

Theorem 1. *Let F be a facet of a compact right-angled hyperbolic four-polytope P. If F has V vertices, then*

$$f \geq 2V + 5. \tag{1}$$

Define the pentagon excess by

$$W = \sum_{R \text{ a ridge of } P} \big(k(R) - 5\big),$$

where $k(R)$ is the number of sides of R. We will prove

$$W = v - 5f, \qquad \mathrm{Vol}(P) = \frac{\pi^2}{12}(f + W + 16). \tag{2}$$

For a facet F, put

$$d(F) = \sum_{R \subset F} \big(k(R) - 5\big).$$

If it has V vertices, then $d(F) = (V - 20)/2$ and $W \geq d(F)$.

Corollary 2. *For every facet F of P with V vertices,*

$$f + W \geq \frac{5}{2}V - 5. \tag{3}$$

Consequently:

1. *if $\mathrm{Vol}(P) < 34\pi^2/3$, then every facet has at most 48 vertices;*

2. *if* $\mathrm{Vol}(P) \le 34\pi^2/3$*, then every facet has at most* 50 *vertices;*

3. *if* $\mathrm{Vol}(P) < 68\pi^2/3$*, then every facet has at most* 104 *vertices.*

The comparison values are respectively the volume of the regular compact right-angled 120-cell and of its facet double. The corollary is a local size reduction. No assertion about minimality, a second volume, or decomposition into regular 120-cells follows from it.

The geometric input is the usual acute-chamber wall property: distinct outward unit normals of an acute-angled hyperbolic polytope have nonpositive scalar products. We use the standard Lorentz model and the ordinary dual cell decomposition of the boundary of a simple convex polytope. These are the acute-angle facts discussed in [2, pp. 63–64 and Proposition 4.1], with the general geometric background in [1]. We prove below the stronger compact normal gap, the intersection and distinctness statements used in the count, and the needed planar closing argument. The volume discussion also uses Chern–Gauss–Bonnet for a compact hyperbolic reflection orbifold; its resulting face formula agrees with [3, Proposition 3.5(d)]. No classification of compact right-angled three- or four-polytopes is an input. In particular we do not assume uniqueness of a small facet or a minimum-volume theorem.

2 Normal geometry and perpendicular edges

Use the hyperboloid model in $\mathbb{R}^{n,1}$, with scalar product of signature $(-,+,\dots,+)$. A point of $\mathbb{H}^n$ is a future unit timelike vector. An outward unit spacelike normal a defines the half-space $\langle x, a\rangle \le 0$.

Lemma 3. *Distinct wall normals of a compact right-angled polytope have product either* 0 *or strictly less than* -1*. A mutually orthogonal collection of wall normals determines an actual common face.*

Proof. Products are nonpositive by the acute-chamber wall property. Suppose two normals a, b have $g = \langle a, b\rangle \in (-1, 0)$. Their Gram matrix

$$G = \begin{pmatrix} 1 & g \\ g & 1 \end{pmatrix}$$

is positive definite and G^{-1} has nonnegative entries. Let p be a strict future timelike interior point and set $\beta = (\langle p, a\rangle, \langle p, b\rangle)^T < 0$. The orthogonal projection

$$u = p - [a\ b]G^{-1}\beta$$

adds nonnegative multiples of the outward normals to p. Thus it strictly satisfies every other wall inequality. It satisfies the two selected equalities, and

$$u^2 = p^2 - \beta^T G^{-1}\beta < 0, \qquad \langle p, u\rangle = u^2 < 0.$$

It is therefore future timelike and gives an actual ridge, whose angle is not $\pi/2$, a contradiction.

If $\langle a, b\rangle = -1$, the nonzero vector $a + b$ is null. It has negative product with p, hence is future. Its products with all other wall normals are nonpositive. It would be a feasible ideal point, contrary to compactness. The sum cannot be zero, because p has negative product with both summands. This proves the gap.

For mutually orthogonal normals $a_1, \dots, a_j$, the same projection is particularly simple:

$$u = p - \sum_{i=1}^{j} \langle p, a_i\rangle a_i.$$

It is future timelike, has zero product with each selected normal and strict negative product with every other wall normal. Hence their intersection is an actual codimension-j face. □

At any finite vertex, all its incident normals lie in a positive definite space. Lemma 3 makes them mutually orthogonal. There are exactly n of them, so the polytope is simple. A facet inherits right angles: its neighboring wall normals are orthogonal to its own normal, and serve as its intrinsic outward normals. The same applies to faces of lower dimension. A compact right-angled k-gon has area $(k-4)\pi/2 > 0$, so $k \geq 5$.

Lemma 4. *Fix a facet A with normal a, and a facet disjoint from A with normal z. Any two common neighbors of these facets meet each other.*

Proof. We have $\langle a, z\rangle < -1$. The plane spanned by a, z is Lorentzian, so its orthogonal complement is positive definite. The normals b, c of two common neighbors belong to this complement. In a positive definite space their product cannot be below -1. Product -1 would imply $b = -c$, incompatible with a strict point of both negative half-spaces. Lemma 3 therefore gives $\langle b, c\rangle = 0$, and the facets meet. □

We call the set of neighbors of A meeting another facet its *footprint on the neighbors of A*. Lemma 4 says that a disjoint facet's footprint is a clique. This is the only meaning of footprint used below.

Lemma 5. *Let A be a facet of a compact right-angled polytope in dimension three or four. For every vertex x of A, the edge leaving A is perpendicular to A and ends on a facet R_x disjoint from A. All the R_x are distinct. Moreover, R_x meets a neighbor B of A if and only if x belongs to B.*

An actual facet disjoint from A meeting all the neighbors of A at x must be R_x.

Proof. Let a be the outward normal of A. Regard x as a vector in $a^\perp$. Simplicity gives exactly one edge at x not contained in A. All its other defining walls are orthogonal to a, so its initial direction is $-a$. If its length is $L_x > 0$, the other endpoint is

$$y = \cosh(L_x)x - \sinh(L_x)a.$$

The new normal there is forced to be

$$r_x = \sinh(L_x)x - \cosh(L_x)a. \tag{4}$$

It is orthogonal to the walls defining the edge, has norm one and satisfies $\langle y, r_x\rangle = 0$, $\langle x, r_x\rangle < 0$. Its product with a is $-\cosh(L_x) < -1$, so R_x is disjoint from A.

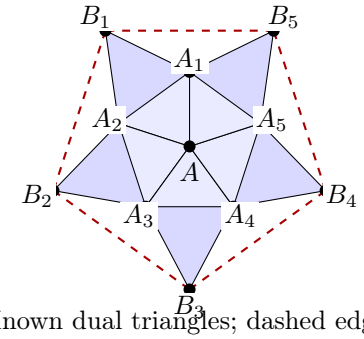
For a neighbor B with normal b,

$$\langle r_x, b\rangle = \sinh(L_x)\langle x, b\rangle.$$

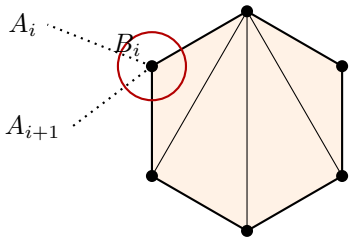
This is zero exactly when $x \in B$; the actual-intersection part of Lemma 3 gives the stated incidence equivalence. If $r_x = r_y$, projection to $a^\perp$ gives $\sinh(L_x)x = \sinh(L_y)y$, hence $x = y$.

Finally a normal orthogonal to all the neighbors defining x lies in $\operatorname{span}(x, a)$. For an actual facet disjoint from A its orientation and its product with x give the form $r = \sinh(L)x - \cosh(L)a$, with $L > 0$. Two distinct normals of this form have product $\cosh(L - L_x) > 0$, contrary to acuteness. Thus $r = r_x$. □

The final assertion is useful: a more distant wall on the same perpendicular line cannot be counted as a second actual facet.



Known dual triangles; dashed edges are forced when there is no extra face.



An ear has two neighbors in the disc and only two outside: total degree four.

Figure 1: The dual closing argument. The left panel depicts the cyclic incidences for $k = 5$; the argument works for every $k \geq 5$. The right panel is a schematic triangulated complementary disc. These are combinatorial drawings, not metric hyperbolic models.

3 Closing the neighborhood of a polygon

Lemma 6. *Let $Q \subset \mathbb{H}^3$ be compact and right-angled, and let A be a k-gonal face. Then Q has at least $2k+2$ faces. If it has exactly $2k+2$, the unique face left after counting A, its k neighbors and its k perpendicular-edge opposites is disjoint from all k neighbors of A.*

Proof. Label the neighbors $A_1, \ldots, A_k$ cyclically. At the vertex where A_i and A_{i+1} meet A, let B_i be the opposite face supplied by Lemma 5. Indices are cyclic. These are $1+2k$ distinct faces. Among the A_j, the only neighbors of B_i are A_i, A_{i+1}. The A_j themselves have just their cyclic adjacencies: any other adjacency, together with A, would give an actual vertex of A by Lemma 3.

Call any further face extra. Such a face meets at most one A_i. Indeed two nonconsecutive A_i cannot both meet it by Lemma 4; meeting two consecutive ones would force it to be the already counted B_i, by Lemma 5.

Pass to the dual triangulated two-sphere. Every vertex has degree at least five, since every face of Q is a compact right-angled polygon. We use the elementary ear fact: a triangulated polygon with no interior vertices has a vertex of degree two within that polygon. If one boundary edge is prescribed, such an ear can be chosen away from its endpoints. For example, root the tree dual to its triangles at that edge and take a leaf away from the root; for a single triangle take the third vertex.

Suppose first there is no extra face. Each A_i has exactly the five known neighbors

$$A,\ A_{i-1},\ A_{i+1},\ B_{i-1},\ B_i.$$

The cyclic link of A_i forces the edge $B_{i-1}B_i$. The triangles around A and the A_i form a disc whose complementary disc has boundary $B_1, \ldots, B_k$ and no interior vertices. An ear in this disc has degree two there, and has only its two outside neighbors A_i, A_{i+1}. Its total degree would be four, a contradiction.

Now suppose there is exactly one extra face C, and it meets A_i. It meets no other A_j. The five-neighbor argument still forces $B_{j-1}B_j$ for $j \neq i$. At A_i its cyclic link instead has the segment B_{i-1}, C, B_i. The complementary disc is therefore a polygon with C inserted into the B-cycle, again without interior vertices. Every B_j has just two outside A-neighbors, so needs degree at least three within this disc. The vertex C has just the outside neighbor A_i, so needs degree at least four there. An ear is impossible. Hence the unique extra face meets none of the A_j, as asserted. □

The equality pattern can be recognized without a small-polyhedron list. When there is exactly one extra face and it meets none of the A_i, it is the unique interior vertex of the complementary

B-disc. A chord between two B_i would cut off a polygon without this interior vertex. The ear fact, applied away from that chord's endpoints, again gives a B_i of total degree four. Thus there are no chords and the extra face joins every B_i. The result has two k-gons and two cyclic rows of k pentagons: the Löbell face lattice. This agrees with the usual description in [4, p. 17 of the author's version]. We need only the proved disjointness of the extra face, and do not invoke metric rigidity of this lattice.

4 Capacity around a three-dimensional facet

Proof of Theorem 1. Let F have V vertices and M two-faces. Trivalence and Euler's formula inside F give

$$M = V/2 + 2. \tag{5}$$

There are M neighboring facets $F_1, \dots, F_M$ and V distinct perpendicular-edge opposites R_x. Thus $1 + M + V$ facets are already counted. Call any other facet extra.

Suppose the ridge $F \cap F_i$ has k_i sides. Inside F_i, the known faces are precisely this ridge, its k_i neighbors coming from the other F_j, and its k_i perpendicular-edge opposites coming from the R_x at its vertices. The perpendicular edges in F_i are the same edges of P used in Lemma 5. If a different F_j meets F_i, their three pairwise orthogonal normals with that of F give an actual common edge by Lemma 3; hence it is already one of those k_i neighbors. The opposite-incidence equivalence in Lemma 5 makes the list of R_x exact as well.

Let e_i count the further faces of F_i, all coming from extra facets of P. Lemma 6 gives $e_i \geq 1$. An extra facet has footprint at most two among the F_j. Indeed its footprint is a clique by Lemma 4. Three members of this clique have mutually orthogonal normals in the Lorentz space $a^\perp$ of signature $(3, 1)$, where a is the normal of F. Together with a, they give an actual vertex x of F. A facet meeting all three is R_x, already counted. A larger orthogonal collection cannot fit in $a^\perp$.

There is an additional restriction when $e_i = 1$. The equality part of Lemma 6 says that its unique extra face is disjoint from all the k_i neighboring faces of $F \cap F_i$ inside F_i. The parent extra facet cannot also meet another F_j: such a meeting would force a common edge with F_i, F_j, contradicting that disjointness. This extra facet therefore has singleton footprint $\{i\}$.

Let s be the number of indices with $e_i = 1$. Their extra facets are distinct and cannot serve any other index. The other $M - s$ indices require at least $\sum_{e_i \geq 2} e_i$ incidences. Since every remaining extra facet serves at most two indices, the number of distinct extras is at least

$$s + \frac{1}{2} \sum_{e_i \geq 2} e_i \;\geq\; s + (M - s) = M.$$

Adding the previously counted facets and using (5) proves

$$f \geq 1 + M + V + M = 2V + 5.$$

□

5 Pentagon excess and the volume reduction

Because P is simple, $e = 2v$. Euler's formula for its boundary gives $v - e + r - f = 0$, hence $r = v + f$. At each vertex the link is a tetrahedron, so six ridges meet there. Counting their corners yields

$$\sum_R k(R) = 6v.$$

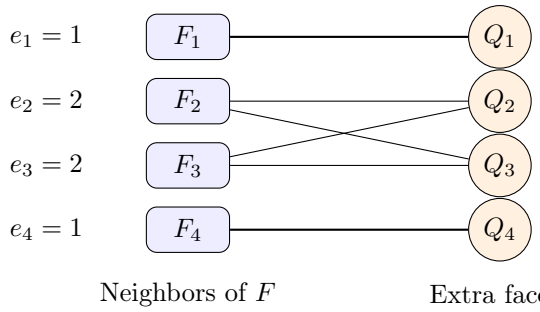


Figure 2: The final capacity count, shown as an incidence graph rather than a claimed polytope realization. An index with $e_i = 1$ consumes a distinct singleton. Every other index needs at least two incidences, while each remaining extra facet can serve at most two indices.

It follows that

$$W = 6v - 5r = v - 5f \geq 0.$$

Inside a facet F, trivalence gives $2E_F = 3V$ and Euler gives $M = V/2 + 2$. Therefore

$$d(F) = 3V - 5M = \frac{V - 20}{2}. \tag{6}$$

Each ridge belongs to two facets, so $\sum_F d(F) = 2W$. In particular $W \geq d(F)$, since all ridge excesses are nonnegative.

For completeness the compact volume formula can be recovered from the reflection orbifold. The stabilizer of the interior of a codimension-j face is a product of j groups of order two. Its Euler characteristic is consequently

$$\chi_{\text{orb}} = 1 - \frac{f}{2} + \frac{r}{4} - \frac{e}{8} + \frac{v}{16} = \frac{v - 4f + 16}{16}.$$

Chern–Gauss–Bonnet in dimension four gives

$$\text{Vol}(P) = \frac{4\pi^2}{3}\chi_{\text{orb}} = \frac{\pi^2}{12}(v - 4f + 16) = \frac{\pi^2}{12}(f + W + 16),$$

as in (2).

Combining Theorem 1 with (6) gives

$$f + W \geq 2V + 5 + \frac{V - 20}{2} = \frac{5}{2}V - 5.$$

If $\text{Vol}(P) < 34\pi^2/3$, then $f + W < 120$, so $V < 50$. The integer V is even by trivalence, giving $V \leq 48$. The non-strict comparison gives $V \leq 50$. Likewise the strict double-volume comparison gives $f + W < 256$, whence $V < 522/5$ and $V \leq 104$. This proves Corollary 2.

6 Limits of the argument

For a dodecahedral facet, Theorem 1 starts at 45 facets. It does not reach 120. Nor does it force a polytope containing a dodecahedral facet to contain a regular 120-cell. The remaining possibilities in the strict smaller-volume problem have facets with at most 48 vertices, but their mutual geometric compatibility is not settled by the single-facet count. The same limitation is stronger still for the proposed second-volume problem.

The proof is a written geometric argument. Its normal, topological and orbifold interfaces have not been formalized in Lean. No numerical search or bounded polytope census is used, and no novelty or published priority claim is attached to the capacity inequality.

References

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